

## Chapter 3

# Euler-Bernoulli beam theory and Deflection of beams

### 3.1 Question Bank UNIT - 3

#### Quiz Questions

1. Relationship between radius of curvature and curvature of the beam is .....
2. When the section modulus increases the bending stress .....
3. The axial force acting on the neutral axis is .....
4. Assumptions of the Euler Bernoulli beam are .....
5. A cantilever beam of rectangular cross-section is subjected to a load  $W$  at its free end. If the depth of the beam is doubled and the load is halved, the deflection of the free end as compared to original deflection will be....
6. In a cantilever beam, if the length is doubled while keeping the cross-section and the concentrated load acting at the free end the same, the deflection at the free end will increase by .....
7. Section modulus of a beam is always given by formula.....
8. Bending occurs due to the application of ..... load
9. In cantilever beam, slope and deflection at free end is .....
10. *Deflection is maximum at a point where slope is zero*, this statement is true for a ..... beam.
11. Moment of resistance is defined as .....
12. The area moment of inertia of a square of size 1 unit about its diagonal is .....
13. The term flexural rigidity can be written as .....
14. What are the conditions to be satisfied for determinate structures.....
15. In a cantilever beam if the length is doubled keeping the cross section and the concentrated load acting at the free end the same the deflection at the free end will increase by .....
16. In case of fixed beam subjected to gravity loads maximum hogging or negative bending moment develops at the .....(supports)

17. What are the units of flexural rigidity?
18. The maximum ..... stresses occurs at top most fibre of a simply supported beam.
19. As compared to uniaxial tension or compression, the strain energy stored in bending is only ..... (1/3)
20. When a rectangular beam is loaded longitudinally shear develops on ..... fibres.

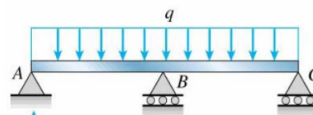
### Main Questions

1. Derive the Differential Equation for the elastic curve of a beam.
2. If the beam cross-section is rectangular having a width of 75 mm, determine the required depth such that maximum bending stress induced in the beam does not exceed  $40 \text{ MN/m}^2$
3. A simply supported beam of rectangular section 4 cm by 6 cm carries a mid-span concentrated load such that the 6 cm side lies parallel to line of action of loading; deflection under the load is . If the beam is now supported with the 4 cm side parallel to line of action of loading, the deflection under the load will be?
4. The deflection curve for a simply supported beam is given by the following equation:

$$W = -\frac{W_o \cdot x}{120LEI}(10L^3 - 10L^2 \cdot x + 5x^2 - x^3) \quad (3.1)$$

Describe the loading on the beam.

5. A beam with a rectangular section of 120 mm 60 mm, designed to be placed vertically is placed horizontally by mistake. If the maximum stress is to be limited, the reduction in load carrying capacity would be?
6. A continuous beam ABC supports a uniform load  $q$ . Determine the reactions. Assume the length between  $AB = BC = L$ .



7. Derive the complete Bending Equation and extend it to an expression for moment carrying capacity of a section.
8. Derive the expression for the deflection of a cantilever beam subjected to a point load at the free end using the double integration method.

9. A circular pipe of external diameter 70mm and thickness 8mm is used as a simply supported beam over an effective span of 2.5m. Find the maximum concentrated load that can be applied at the centre of the span if the permissible stress in the tube is  $10\text{N/mm}^2$
10. A simply supported beam, 2 in wide by 4 in high and 12 ft long is subjected to a concentrated load of 2000 lb at a point 3 ft from one of the supports. Determine the maximum fiber stress and the stress in a fiber located 0.5 in from the top of the beam at midspan.
11. Determine the maximum value of  $EI_y$  for the cantilever beam loaded as shown in Fig. Take the origin at the wall.

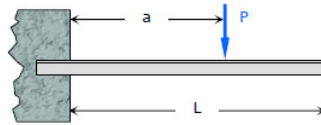


Figure P-607

12. Derive the expressions for beams subjected to combined loads inclined loads and eccentric loads.